

Arkime (formerly Moloch) – Notes

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Purpose: Fast search and analysis of full network traffic (PCAPs) for hunting C2, phishing, exfil, and malware.

Main Interface:

- **Sessions tab:** Main view of all traffic sessions.
- **SPI View:** Expand a session for full details (headers, body, MD5).
- **Connections tab:** Visual graph of IP relationships.
- **Hunt tab:** Saved keyword searches across historical data.

Basic Workflow:

1. Ingest PCAP (script clears old data, captures, restarts viewer).
2. Adjust time range (top left: All or custom dates).
3. Click any value (port, IP, URI) to add as filter.
4. Use negation (and not) to remove known-good traffic.
5. Expand session (left icon) for deep details.

Analyzing Phishing Traffic

What to find: Malicious downloads via HTTP (few headers, disguised files).

Steps:

1. Click destination port **80** → add as filter (HTTP traffic).
2. In Information column, click URI → and not to remove known-good domains.
3. Expand suspicious session → check HTTP request (few headers = scripted).
4. Look at response: large databytes claiming PDF but showing PE (MZ header or "This program cannot be run in DOS mode").
5. Click **Show Images & Files** → download payload.
6. Use body MD5 for VirusTotal search.

Why: Minimal headers + disguised executables on port 80 is a common initial access method.

Detecting Malware Use of TLS Connections

What to find: Malware hiding C2 or exfil in TLS (often non-standard ports).

Steps:

1. Filter protocol to **TLS** + source IP.
2. Expand session → check certificate (self-signed or suspicious subject).
3. Note **JA3** hash → click to filter all sessions using same JA3.
4. Use Connections graph to see related IPs.

Why: JA3 fingerprints the TLS handshake even when data is encrypted, linking suspicious sessions.

Developing Techniques for Detecting Data Exfiltration

What to find: Stolen data leaving via SMTP, HTTP POST, or zipped files.

Steps:

1. Go to **Hunt** tab → Create search.
2. Search for keywords: "system info", "password", "cookie", "bank".
3. Set to source packets only (outbound exfil).
4. Expand matching session → **Show Images & Files** to download attachment.
5. For Base64: save content → decode with `cat file.b64 | base64 -d > output.jpg`.

Why: Hunt searches historical data for patterns; decoding reveals exactly what was stolen.

Quick Arkime Tips

- Click any field to filter (fastest way).
- Use **Connections** graph to see relationships.
- Check **body MD5** in SPI view for instant VT lookup.
- Export files directly from **Show Images & Files**.